

BMW Group Standard

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Replacement for
GS 90010-1:2021-03

Type of surface protection for metallic materials

Metallic coatings and inorganic coating systems

Standard parts and standard-related drawing parts

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Foreword

The German version is binding.

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This Group Standard has been coordinated with the responsible departments of the BMW Group. The names of the persons creating, releasing and checking this Group Standard are filed and can be retrieved in the master data of the TERE system.

Amendments

The following amendments have been made to GS 90010-1:2021-03:

- section 6.2, section 6.3 and section 6.4 updated;
- section 7 is now section 4;
- Table 1: requirement KWT according to DIN EN ISO 11997-1 cycle B at ZNNID SW and ZNNID SI added (was footnote);
- Table 1: column "Notes and Remarks" changed;
- section 8.4 changed;
- section 9 added, was GS 90010-1 BBL;
- cited standards updated;
- Group Standard editorially revised.

Previous editions

BMWN(S) 60000.0: 1998-11

GS 90010: 2001-07, 2003-10, 2004-02, 2008-02

GS 90010-1: 2010-02, 2011-03, 2020-01, 2021-03

GS 90010-1 BBL1: 2014-06, 2021-03

1 Scope and purpose

This Group Standard is applicable for inorganic coating systems on metallic materials. In order to assure the corrosion protection in case of Duplex systems (ZNNID SI, ZNNID SW) additional inorganic protective layers (topcoats) are required.

This Group Standard defines the coating system requirements for standard and standard-similar drawing parts and the corresponding tests.

This Group Standard is not applicable for:

- structural components, see GS 90010-2;
- metal parts with organic coating, refer to GS 90011;
- surfaces or components with mainly decorative purpose;
- decor and accent surfaces in interior.

2 Normative references

This Group Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

DIN 50969-1	Prevention of hydrogen-induced brittle fracture of high-strength steel building elements; Part 1: Advice on the prevention
DIN 50969-2	Prevention of hydrogen-induced brittle fracture of high-strength steel building elements; Part 2: Test methods
DIN EN ISO 1456	Metallic and other inorganic coatings; Electrodeposited coatings of nickel, nickel plus chromium, copper plus nickel and of copper plus nickel plus chromium
DIN EN ISO 1463	Metallic and oxide coatings; Measurement of coating thickness; Microscopical method
DIN EN ISO 4042	Fasteners; Electroplated coating systems
DIN EN ISO 4527	Metallic coatings; Autocatalytic (electroless) nickel-phosphorus alloy coatings; Specifications and test methods
DIN EN ISO 6270-2	Paints and varnishes; Determination of resistance to humidity; Part 2: Condensation (in-cabinet exposure with heated water reservoir)
DIN EN ISO 9227	Corrosion tests in artificial atmospheres; Salt spray tests
DIN EN ISO 9717	Metallic and other inorganic coatings; Phosphate conversion coating of metals
DIN EN ISO 10683	Fasteners; Non-electrolytically applied zinc flake coating systems
DIN EN ISO 11997-1	Paints and varnishes; Determination of resistance to cyclic corrosion conditions; Part 1: Wet (salt fog)/dry/humid
DIN EN ISO 15330	Fasteners; Preloading test for the detection of hydrogen embrittlement; Parallel bearing surface method
DIN EN ISO 16047	Fasteners; Torque/clamp force testing
GS 90003-1	Tightening torques and clamping forces for screws and nuts with metric thread; Fundamentals and calculation procedures
GS 90003-2	Tightening torques and clamping forces for screws and nuts with metric thread made of steel and aluminium
GS 90011	Coating of parts made of metallic materials by means of organic materials; Requirements and tests
GS 91005-1	Technical drawings; Drawing generation, design rules, agreements
GS 92001	Bolts, screws, studs and nuts; Designations, types and designs
VDA 235-101	Friction coefficient setting of mechanical fasteners with metric thread
VDA 235-104	Cr(VI)-free surface protection coating systems for fasteners with metric thread
VDA Band 2	Quality management in the automotive industry; Securing the quality of supplies; Production process and product approval (PPA)

3 Abbreviations, terms, definitions and symbols

3.1 Abbreviations

There are no abbreviations in this document.

3.2 Terms

Surface protection	Surface protection is achieved by chemical/electro-chemical or physical application of coatings or by changing the component surface into a corrosion or wear resistant condition. Apart from possible decorative effects, the main objective of surface treatment is the achievement of corrosion protection adapted to stress and requirements.
Lubricant	Lubricants in the sense of this Group Standard are substances applied additionally for the optimization/setting of friction values.

3.3 Definitions

There are no definitions in this document.

3.4 Symbols

D	nominal thread diameter nut
d	nominal thread diameter screw
R_m	tensile strength
μ_b	under head friction
μ_{th}	thread friction
μ_{tot}	total friction

4 Designation

4.1 General

Type of surface protection may only be entered on drawings and other documents with BMW code.

4.2 Description of standard parts

Descriptions according to the respective product standards and BMW code e.g. ZNS3.

EXAMPLE Hexagon screw ISO 4014 - M10x80-8.8-ZNS3

4.3 Drawing entry for drawing parts similar to standard

Drawing entry in accordance with GS 91005-1.

Description of surface protection according to GS 90010-1 with BMW code ZNNIV SI:

GS 90010-1 - ZNNIV SI

Non-standardized surface protection systems shall be specified in the drawing after consultation with the specialist department "Water Induced Total Vehicle Strain, Corrosion Resistance", stating the exact layer designation and a suitable test.

5 General

5.1 Fasteners

Fasteners in this Group Standard are for example:

- screws and screws with captive, easily turning washers;
- nuts and nuts with captive, easily turning washers;
- self-pierce rivets, blind rivets, weld studs.

NOTE The surface protection types ZNS2 and ZNS3 are not suitable for nuts with non-metallic clamping parts.

5.2 Hydrogen-induced brittle failure

Hydrogen induced brittle failure may occur on high strength steel parts e.g. due to pickling or electroplating. Therefore, safety-relevant, high stressed parts with a tensile strength of $R_m \geq 1\,000\text{ MPa}$ should not be electroplated.

In such cases

- dimensioning shall be such that materials with lower strength can be used alternately,
- a material shall be specified that is not susceptible to hydrogen induced brittle failure,
- a material shall be selected that is durable without surface treatment or
- a different coating process shall be selected in agreement with material development.

Examples of such processes are as follows:

- special processes (ZNS2, ZNS3);
- mechanical plating;
- thermo-chemical treatment;
- organic coatings.

The parts shall always be cleaned with mechanical or alkaline methods. If chemical cleaning is required due to functional reasons only acids with suitable inhibitors shall be used for the preparation/cleaning.

In case of exceptions, which are subject to agreement with material development, measures shall be taken in accordance with DIN EN ISO 4042 and DIN 50969-1 to avoid hydrogen-induced brittle fractures. These shall be documented in the initial sample inspection report.

5.3 Layer thicknesses

The values specified in the tables are recommended minimum layer thicknesses. The release depends on the compliance with the respective corrosion protection requirements; consequently, the required layer thickness can be higher than the specified recommended minimum layer thicknesses (see section 7).

During the coating process the manufacturer shall ensure that the finished dimension of the parts is within the permitted tolerances (see section 8.5).

As the coated thread shall be within the tolerances specified for the component, the tolerance position of the uncoated thread and the thickness of the protective surface coating are required to match.

Processes for the large-scale application of zinc flake coatings often require compromises in the uniformity of coating thicknesses.

Depending on the geometry of the threaded parts, different layer thicknesses can be achieved on parts of the surface of one and the same threaded part.

If a uniform, dimensionally accurate layer thickness is achieved on the predominant part of the total surface, material accumulations can occur predominantly in the thread base and on the contact surface (e.g. under the screw head), which exceed the usually specified tolerance limits, but do not necessarily mean that the threaded parts cannot be used.

In such cases, the dimensional accuracy of coated threaded parts can no longer be determined by means of gages, but only by direct measurement.

When gaging coated threads of screws a maximum screw-on torque of $0,001d^3$ (Nm) over a length of $1d$, starting from the thread end, is permitted. When gaging coated threads of screws a maximum screw-in torque of $0,001D^3$ (Nm) is permitted. d/D is the nominal thread diameter (screw/nut) in mm.

For zinc flake coatings and especially for coatings with an additional organic top layer, it shall be taken into account that possible accumulations of material on the contact surfaces (e.g. under the screw head) of threaded parts can lead to setting.

This can be tolerated to the extent that the setting does not have excessive negative effects. There are no reliable limit values; experience to date has shown that a maximum value of $25\text{ }\mu\text{m}$ is still acceptable for material accumulations under contact surfaces of threaded parts.

If, in justified cases, certain layer thicknesses may not be exceeded, the maximum permissible layer thickness shall be specified in the drawing.

5.4 Frictional behavior

5.4.1 Fasteners with metric thread

For fasteners with metric thread, the requirements of GS 90003-1, GS 90003-2, VDA 235-101 and DIN EN ISO 16047 are applicable.

The friction coefficients are also applicable for screw and washer assemblies according to GS 92001.

These fasteners shall be generally adjusted for a friction coefficient range of $\mu_{\text{tot}} = 0,09$ to $0,15$.

Including measurement uncertainties, the value of the partial friction coefficients (μ_{b} , μ_{th}) may not fall below $0,08$ or exceed $0,16$.

For a series of measurements, a mean value of $\mu_{\text{tot}} = 0,12$ shall be targeted.

The requirements are also applicable for fasteners without surface protection.

The required friction coefficients can also be achieved by use of a lubricant integrated in the coating system or an additionally applied lubricant.

If lubricants are used, these shall be specified in the drawing/initial sample inspection report.

The lubricant can be colored to prove that an additional lubricant has been applied to silver-colored coatings (see VDA 235-101).

The influence of lubricants on bonding, clamping and sealing screw locking devices (e.g. microencapsulation) shall be taken into account. The compatibility of lubricants applied on plastics shall be considered accordingly.

For parts without thread, the surface protection types specified in this Group Standard may be used without lubricant coating.

5.4.2 Fasteners with non-metric thread

The respective friction coefficients shall be determined together with the respective development partner by measurement in each case.

This applies to sheet-metal (self-tapping) and plastic screws. If lubricants are used, these shall be indicated in the drawing/initial sample inspection report.

5.5 Adhesion promoter

For the use of adhesion promoters (e.g. phosphatizing, passivation), system compatibility with subsequent sealing or top coats as well as freedom from CR(VI) shall be guaranteed.

5.6 Contact corrosion

To prevent contact corrosion in combination with aluminum components, either ZNNIV SI, ZNNIV SW, ZNS3, ZNNID SW coatings or ZNNID SI coatings shall be used. For the body-in-white, ZNNI SI and ZNS2 are also permitted.

In case of contact with magnesium, stainless steel and carbon fiber reinforced plastics (CFRP), the selection of the type of surface protection is subject to agreement with the specialist department "Water Induced Total Vehicle Strain, Corrosion Resistance".

5.7 Electrically conductive joints

For electrically conductive connects SNZN 70/30 shall be preferred. Other coatings shall be harmonized with specialist department „Water Induced Total Vehicle Strain, Corrosion Resistance“.

5.8 Coloration

Coloring may vary considerable depending on the surface. Additional coloring in lubricants is permitted only in agreement with the releasing specialist department "Water Induced Total Vehicle Strain, Corrosion Resistance".

5.9 Contamination

Contamination resulting from auxiliary means such as oils and grease, e.g. when rolling-on washers on already coated components is not permitted.

5.10 Micro-encapsulated and clamping coatings

For micro-encapsulated (MKL) and clamping coating systems (KL) that ensure adhesion according to DIN 267-27 and DIN 267-28 (e.g. ZNNI system without topcoat) shall be selected.

Threaded parts with thread locking and sealing materials are generally excluded from an additional treatment with lubricants according to VDA 235-101. Friction coefficient requirements according to section 5.4.1 cannot be fulfilled.

6 Types of surface protection, brief description

6.1 Phosphate coatings (PHR, PHR OIL)

Phosphate coatings are defined in accordance with the procedure described in DIN EN ISO 9717.

Phosphate coatings (PHR) have a low temporary corrosion protection effect. They are used mainly as adhesion promoters for organic coating materials.

Oiled phosphate coatings (PHR OIL) provide temporary corrosion protection for shipment and storage. Friction coefficient requirements according to section 5.4.1 cannot be fulfilled.

6.2 Electroplated zinc coatings (A5, ZNT, ZNFE SW, ZNFE SI, ZNNI SI, ZNNIV SI, ZNNIV SW)

Zinc coatings, transparent passivated (ZNT), without passivation (A5).

Low alloy zinc coatings, black or transparent passivated and sealed (ZNFE SW, ZNFE SI). These zinc coatings contain small amounts of iron (approximately 0,3 % to 1 %) and/or cobalt (approximately 0,6 % to 1,2 %) as alloying elements.

Zinc coatings, high-alloy, transparently, passivated (ZNNI SI) contain 10 % to 15 % nickel.

High alloy zinc coatings, transparent (ZNNIV SI) or black (ZNNIV SW) passivated and sealed contain 10 % to 15 % nickel.

For components which are subjected to a downstream paint process the use of any type of sealing is not allowed. If passivations are used, phosphatability of components in conjunction with the painting process shall be verified.

In case of black passivation, solutions containing silver are not permitted.

6.3 Electroplated zinc coatings with organic top coat (ZNNID SW, ZNNID SI)

High alloy zinc coatings (10 % to 15 % nickel) with additional silver (ZNNID SI) or black (ZNNID SW) organic topcoat contain 10 % to 15 % nickel.

The multilayer coating shall be applied in a continuous coating process.

6.4 Zinc flake coatings (ZNS3, ZNS2)

Zinc flake coatings (ZNS3 base und topcoat) contain mainly aluminum tinsel and up to 10 % aluminum tinsel or alloying tinsel with aluminum and/or magnesium (particle size approximately $0,25 \mu\text{m} \pm 0,10 \mu\text{m}$) in inorganic or organic suspension. These are stoved for hardening. The stoving temperatures (min./max.) and stoving times specified by the chemical manufacturer shall be observed.

To assure coating adhesion, residues and deposits (e.g. parting agents, scale, silicates, rolled-in particles) from previous manufacturing processes shall be completely removed by suitable methods (e.g. pickling, peening) so that they do not impair coating adhesion in any way.

In cured condition, these coating shall be made up of mainly metallic constituents.

As a prerequisite for a cathodic protection effect the coating shall be electrically conductive. To prevent high settling rates and to achieve a good barrier layer, the metal particles in the coatings shall be aligned mainly parallel to the surface.

Zinc flake coatings have a low ductility. Therefore, these coatings are not allowed for components that are deformed or pressed-in after the coating process.

In case of mechanical fasteners with captive washer, sticking may occur depending on the geometry.

Unless specified otherwise, threaded parts that were rolled to increase fatigue strength, coating methods shall be chosen in which stoving temperature does not exceed 220 °C. The same restriction applies for surface hardened threaded parts.

Zinc flake coatings without topcoat (ZNS2) are only permitted for used in body shop.

6.5 Electroplated nickel coatings (NI)

Single layer coating with low corrosion protection.

6.6 Chemical nickel coatings (NIH)

Hard nickel (phosphorus) coatings with moderate corrosion and wear protection.

6.7 Electroplated tin coatings (SN)

Tin coatings do not have an anodic protection effect on steel. Unsuitable for use in open air.

6.8 Electroplated tin zinc alloy coatings (SNZN 70/30)

High-alloy tin coating with 65 % to 75 % tin.

6.9 Electroplated copper coatings (CUG)

Copper coatings shall generally be used as intermediate layer. In agreement with the specialist department "Water Induced Total Vehicle Strain, Corrosion Resistance", copper coatings can be used as direct corrosion protection surface.

7 Surface protection types, characteristics and requirements

7.1 Requirements for high strength parts

For parts with a tensile strength of $R_m \geq 1\,000\text{ MPa}$ the surface and heat treatment processes shall comply with state-of-the-art technology for prevention of hydrogen induced brittle failures.

This applies particularly for electroplated coatings. Even with zinc flake coatings, the parts may not be pickled in acids without inhibitors during pretreatment.

The time between pretreatment and coating shall be as short as possible.

Proof of process reliability can be provided by means of stress test according to DIN EN ISO 15330 or DIN 50969-2. This shall be documented in the initial sample inspection report.

Electroplated zinc and zinc alloy coatings are not permitted for parts with a tensile strength of $R_m \geq 1\,200\text{ MPa}$.

7.2 Corrosion protection requirements

7.2.1 General

In case of surface protection types according to Table 1 coatings with additional applied lubricants shall comply with the required corrosion protection characteristics both in lubricated as-delivered conditions and in degreased condition.

7.2.2 Electroplated zinc, zinc alloy and zinc flake coatings

Table 1: Characteristics

BMW code	Appearance	Temperature resistance max. °C	Layer thickness min. µm	NSS¹⁾ according to DIN EN ISO 9227 h	Notes and Remarks
Only in body shop prior to painting as well as assembly components with subsequent paint coating					
A5²⁾	silver	-	5 max. 10	48	Designation according to: DIN EN ISO 4042 - Zn5/U/T0/48h
ZNNI SI³⁾	silver	-	6	480	Designation according to: DIN EN ISO 4042 - ZnNi/An/T0/480h Sealing is not permitted. Passivations shall be validated in BMW target plant.
ZNS2³⁾	silver	-	8	480	Topcoat not permitted.
ZNT	silver	-	5	240	Designation according to: DIN EN ISO 4042 – Zn/An/T0/240h Sealing is not permitted. Passivations shall be validated in BMW target plant.
Passenger compartment, luggage compartment					
ZNT^{4) 5)}	silver	90	5	240	Designation according to: DIN EN ISO 4042 – Zn/An/T0/240h VDA 235-104 - OfI 20
ZNFE SW^{3) 5) 6)}	black	120	5	120 ^{7) 8)} 480 ⁸⁾	Designation according to: DIN EN ISO 4042 - ZnFe/Fn/T2/480h
ZNFE SI^{3) 5) 6)}	silver	120	5	480 ⁸⁾	Designation according to: DIN EN ISO 4042 - ZnFe/An/T2/480h

BMW code	Appearance	Temperature resistance max. °C	Layer thickness min. µm	NSS ¹⁾ according to DIN EN ISO 9227 h	Notes and Remarks
Vehicle exterior, vehicle underside, engine compartment					
ZNNI SI³⁾	silver	180	6	720	Designation according to: DIN EN ISO 4042 - ZnNi/An/T0/720h
ZNNIV SI⁶⁾	silver	180	6	720 ⁸⁾	Designation according to: DIN EN ISO 4042 - ZnNi/An/T2/720h
ZNNIV SW⁶⁾	black	180	6	240 ^{7) 8)} 720 ⁸⁾	Designation according to: DIN EN ISO 4042 - ZnNi/Fn/T2/720h
ZNS3⁹⁾	silver	180 ¹⁰⁾	8	720	DIN EN ISO 10683 - fZn/nc/TL/720h VDA 235-104 - OfI 42
ZNNID SW³⁾	black	130	6 + 6	10 cycles alternating climate test according to DIN EN ISO 11997-1 cycle B ^{1), 7)}	electroplating ZnNi + organic top coat
ZNNID SI³⁾	silver	130	6 + 6	10 cycles alternating climate test according to DIN EN ISO 11997-1 cycle B ^{1), 7)}	electroplating ZnNi + organic top coat

- 1) Base metal corrosion (red rust) is not permitted.
- 2) Permitted only for parts subjected to resistance welding.
- 3) Not suitable for threaded parts in the respective scope of application. The use shall be tested by means of friction coefficient and bolting tests.
- 4) Seam sealing with system-related layer thickness between 0,5 µm and 2,0 µm is permitted and shall be indicated in the initial sample inspection report.
- 5) For use in directly visible area only by agreement with the specialist department "Water Induced Total Vehicle Strain, Corrosion Resistance".
- 6) Plus sealing with system-related layer thickness between 0,5 µm and 2,0 µm.
- 7) Zinc corrosion (white rust) is not permitted.
- 8) Test after heat treatment of 24 h at 120 °C.
- 9) For parts with thread > M6.
- 10) In case of application temperatures higher than 130 °C the parts shall be subjected to 96 h at 180 °C prior to the corrosion test.

7.2.3 Electroplated coatings made of nickel and nickel-phosphate alloys

Table 2: Characteristics

BMW code	Appearance	Layer thickness min. µm	NSS ¹⁾ according to DIN EN ISO 9227 h	Notes and Remarks
Exterior and interior				
NI	silver	10	24	DIN EN ISO 1456 - Fe/Ni10p
NIH	silver	20	240	DIN EN ISO 4527 - Fe/NiP(20)
1) Base metal corrosion (red rust) is not permitted.				

7.2.4 Electroplated tin and copper coatings

Table 3: Characteristics

BMW code	Appearance	Layer thickness min. µm	NSS ¹⁾ according to DIN EN ISO 9227 h	Notes and Remarks
Exterior and interior				
SN	silver	8	144	-
SNZN 70/30	silver	10	480	-
CUG	copper colored	12	-	-
1) Base metal corrosion (red rust) is not permitted.				

7.2.5 Phosphate coatings

Table 4: Characteristics

BMW code	Appearance	Layer thickness min. µm	Test
PHR OIL¹⁾	light grey to dark grey	1	24 h ²⁾ DIN EN ISO 6270-2 CH
PHR¹⁾	light grey to dark grey	1	-
1) Minimum value of overall coefficient of friction $\mu_{\text{tot}} = 0,08$. Only temporary protection during storage. Use permitted only within oiled units compartments.			
2) Test for base material corrosion (red rust).			

8 Test procedure

8.1 Acceptance and test specification

The test includes appearance and general requirements, layer thicknesses, dimensional tolerances as well as climate and corrosion resistance.

8.2 Initial sample inspection

Release of initial samples, which the manufacturer shall submit in sufficient quantity and in good time, is a prerequisite for initial delivery. The test shall be carried out according to VDA Band 2.

8.3 Appearance and general requirements

The coatings shall not have any defects that may impair the appearance and/or the function. Parts from series supply shall correspond to the initial sample in terms of gloss, decorative appearance and corrosion resistance.

8.4 Friction coefficients

Friction coefficients shall be determined according to DIN EN ISO 16047 with the following reference parts:

- reference contact surface HH, bright, uncoated, degreased;
- reference screw with thread tolerance 6 g, bright, uncoated, degreased;
- reference nut with thread tolerance 6 H, bright, uncoated, degreased.

The reference parts shall be cleaned with one of the solvents isohexan, acetone or nitro thinner as specified in DIN EN ISO 16047. Cleaning agents with lubricating or preserving effect are not permitted.

The friction diameter shall be determined and calculated by measurement of the contact pattern according to VDA 235-101 in deviation to DIN EN ISO 16047. The contact patterns shall be documented.

The friction coefficients for total friction μ_{tot} , thread friction μ_{th} and under head friction μ_b shall be determined on 10 parts each of one supply batch. For compliance with the requirements, only one value of the partial friction coefficients may lie outside the ranges specified in section 5.4.1, provided that total friction is within the specifications according to section 5.4.1.

Partial friction and total friction shall be indicated in the friction coefficient protocol with up to 3 decimal places. The evaluation shall be rounded up mathematically to 2 decimal places.

Fasteners with metric thread are excluded from the friction coefficient test.

8.5 Layer thicknesses

The layer thickness on screws is determined on the screw head (refer to DIN EN ISO 4042 and DIN EN ISO 10683). The measuring procedure shall be selected as appropriate. In case of doubt, the metallographic analysis on the longitudinal section according to DIN EN ISO 1463 applies. DIN EN ISO 1463 shall apply.

Layer thicknesses shall be adapted to the respective tolerance positions of the threaded parts. A significant increase in layer thicknesses in the thread compared of the specified minimum layer thicknesses requires the corresponding dimensions of bright threaded parts (see DIN EN ISO 4042 und DIN EN ISO 10683).

8.6 Adhesion

8.6.1 General

Adequate adhesion of the protective surface coatings is assured with properly applied coatings and is therefore not required as an acceptance criterion. In case of doubt, the adhesive strength can be determined as follows.

8.6.2 Electrolytic coatings

The test specimens are stored 30 min at 220 °C and are then quenched in water with room temperature. The coating may not show any blisters or peeling.

8.6.3 Zinc flake coatings and electroplated coatings with organic top coat

Textile adhesive tape with a bonding strength of (7 ± 1) N per 25 mm width is manually pressed-on and then pulled off jerkily. It shall not be possible to peel off the coatings, slight adhesion of the coating material on the pulled off adhesive tape is acceptable.

8.6.4 After coating of the deformed components

In case of parts that are formed after coating, the protective coatings shall be such that they are not separated from the basic material or from each other and that they do not crack.

8.7 Corrosion tests

Passivated or passivated and sealed (topcoated) parts that are subjected to a salt spray mist test NSS according to DIN EN ISO 9227 are heat treated prior to the test to take into account stress during operation or possible impairments in case of repainting.

As regards the analysis of the corrosion protection of the coatings of screws and nuts the surfaces visible after assembly are decisive. Threads, shank sections etc. shall not be included in the analysis. In case of deposits of corrosion products of neighboring areas these are to be excluded from the evaluation.

Corrosion resistance of parts from series deliveries shall correspond to the initial sample.

The parts are considered to be defective if basic material/zinc corrosion occurs before the end of the prescribed test period or if the parts do not correspond to initial sample quality in terms of appearance and corrosion resistance.

8.8 Continuity in series

If the type of surface protection layer in question permits alternatives, the supplier may select such an alternative, (see section 9). The coater and the type of finish selected and sampled shall be communicated to the client, e.g.:

- phosphating process and aftertreatment;
- in case of electroplated alloy type coatings with specification of alloy element concentrations if necessary;
- type of coating material.

The sampled finish of the surface protection is binding for the subsequent series and may only be changed with the consent of the client. This also applies to the change of the coater and the coating process. In these cases, a new initial sample inspection is required.

8.9 Other tests

Further test procedures and cycles shall be coordinated with the specialist department "Water Induced Total Vehicle Strain, Corrosion Resistance" and noted on drawings or quality specifications.

The assessment shall be based on the specifications in section 8.7.

9 Coating systems

In case of fasteners with metric thread and zinc flake coating ZNS3 one of the following CrVI-free coating systems shall be used:

- GEOMET® 321 + PLUS® VLh;
- GEOMET® 321 + PLUS® VL;
- DELTA-PROTEKT® KL 100 + VH 301.1 GZ;
- DELTA-PROTEKT® KL 100 + VH 301 GZ;
- DELTA-PROTEKT® KL 120 + VH 301.1 GZ.

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An additional application of a lubricant to achieve the required friction coefficient is subject to approval of the specialist department "Connection Methods".

NOTE In exceptional cases, other coating systems that verifiably meet the requirements of this Group Standard may also be used in agreement with the specialist department "Water Induced Total Vehicle Strain, Corrosion Resistance". These coating systems shall be specified in the drawing in addition to the code. All coating systems applied shall be listed in the initial sampling inspection report (upon request also confidentially).